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# Feature Representation, Cluster Structure and Explanation Tradeoffs in Customer Segmentation

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# Contents

|                                                         |    |
|---------------------------------------------------------|----|
| Abstract                                                | 6  |
| Declaration                                             | 7  |
| Intellectual Property Statement                         | 8  |
| AI Statement                                            | 9  |
| 1 Introduction                                          | 10 |
| 1.1 Research background . . . . .                       | 10 |
| 1.2 Research questions and contribution . . . . .       | 11 |
| 1.3 Dissertation structure . . . . .                    | 11 |
| 2 Literature Review                                     | 12 |
| 2.1 Representation in customer segmentation . . . . .   | 12 |
| 2.2 Cluster structure and validation . . . . .          | 12 |
| 2.3 Explainable clustering . . . . .                    | 13 |
| 2.4 Explanation complexity and fidelity . . . . .       | 13 |
| 2.5 Research gap and positioning . . . . .              | 14 |
| 3 Data and Methodology                                  | 15 |
| 3.1 Research design . . . . .                           | 15 |
| 3.1.1 Data source . . . . .                             | 15 |
| 3.1.2 Feature engineering . . . . .                     | 16 |
| 3.2 Analytical methods and mathematical scope . . . . . | 17 |
| 3.2.1 Clustering candidates and selection . . . . .     | 18 |
| 3.2.2 Resampling and explanation evaluation . . . . .   | 20 |

|       |                                                               |    |
|-------|---------------------------------------------------------------|----|
| 3.2.3 | Projection filtering and complexity . . . . .                 | 21 |
| 3.2.4 | A local stability certificate . . . . .                       | 22 |
| 3.2.5 | Benchmark and robustness designs . . . . .                    | 24 |
| 4     | Results and Analysis                                          | 26 |
| 4.1   | Algorithm benchmark . . . . .                                 | 26 |
| 4.2   | Representation and customer cluster structure . . . . .       | 26 |
| 4.3   | Explanation complexity, fidelity and addressability . . . . . | 29 |
| 4.4   | Sensitivity and robustness . . . . .                          | 30 |
| 4.4.1 | Named features and reduced geometry . . . . .                 | 30 |
| 4.4.2 | Perturbation and temporal assumptions . . . . .               | 31 |
| 4.4.3 | Dimensionality control . . . . .                              | 32 |
| 4.5   | Integrated discussion and validity limits . . . . .           | 32 |
| 5     | Conclusion                                                    | 34 |
|       | References                                                    | 35 |
| A     | Technical Reproduction Map                                    | 38 |

# List of Tables

|     |                                                                             |    |
|-----|-----------------------------------------------------------------------------|----|
| 2.1 | Comparison of explanation designs in the reviewed literature . . . . .      | 13 |
| 3.1 | Recorded feature contract for the two customer representations . . . . .    | 16 |
| 4.1 | Highest external ARI after label free selection on each benchmark dataset . | 27 |
| 4.2 | Deployment acceptable clustering results on the shared purchaser cohort . . | 27 |

# List of Figures

|     |                                                                                            |    |
|-----|--------------------------------------------------------------------------------------------|----|
| 4.1 | Internal structure and resampling stability for the selected customer partitions . . . . . | 27 |
| 4.2 | Held out ExKMC fidelity across the fixed leaf budget . . . . .                             | 29 |
| 4.3 | Reselected feature deletion and PCA diagnostics . . . . .                                  | 31 |

# Abstract

Customer segmentation groups people using measured attributes, so the choice of those attributes may change both the groups and their explanations. This dissertation compares purchase summaries with a richer set of behavioural features for the same 11,719 RetailRocket purchasers. It applies K Means and CLASSIX under a shared evaluation design, then examines the resulting assignments, their repeatability, and the information needed to describe them. CLASSIX provides a record of how its groups were formed; ExKMC provides decision rules that approximate K Means assignments.

The two representations produce substantially different customer groups, even though repeated fitting within each representation generally gives similar assignments. Adding behavioural features also changes how accurately short decision rules reproduce the groups and how detailed the CLASSIX construction record becomes. Only one selected cluster meets the study's operational checks for size, data coverage and distinctive features. Sensitivity analyses qualify these findings under changes to features, numerical inputs and session definitions.

Representation is therefore part of the clustering model rather than a preliminary choice without consequences. In this dataset, richer information changes the grouping and its explanation, but does not consistently improve geometric quality or yield groups ready for operational use. Human understanding, campaign effectiveness and realised business value remain unmeasured.

# Declaration

This report is my original work unless referenced clearly to the contrary. No portion of the work referred to in this report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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# AI Statement

Artificial intelligence tools were used as assistants for code drafting, data processing, numerical checks, source discovery and English language polishing. The author determined the research design, verified all sources and results, reviewed every substantive claim and accepts responsibility for the report. Details relevant to reproducibility are recorded in the accompanying materials.

# Chapter 1

## Introduction

### 1.1 Research background

Customer segmentation groups people by recorded characteristics, including purchases, browsing and other interactions. Customers need not be purchasers; this study concerns only visitors with a recorded purchase in RetailRocket.

A clustering algorithm receives numerical feature vectors. The *representation* is the choice and construction of these features; their *geometry* consists of distances and neighbourhood relationships between observations. Purchase summaries and additional browsing or category measures can make the same people appear similar in different ways. Comparing algorithms without controlling these choices can obscure why partitions differ.

A *partition* assigns observations to groups, allowing a separate noise state where a method leaves observations unassigned. Clusterability concerns whether data contain structure suitable for clustering, but no definition applies to every structure (Ackerman and Ben-David, 2009; Adolfsson, Ackerman and Brownstein, 2019). The present study therefore examines observed grouping under a specified representation and evaluation design.

Explanation introduces a second distinction. CLASSIX forms and merges local groups; its *provenance* records the route from an observation through those groups to its final cluster (Chen and Güttel, 2024). K Means assigns observations to centres, while ExKMC subsequently builds threshold rules approximating those assignments (Frost, Moshkovitz and Rashtchian, 2020). A construction record and a rule approximation answer different questions; neither alone establishes usefulness or human understanding.

## 1.2 Research questions and contribution

The comparison holds the 11,719 purchasers, event snapshot, common purchase variables, candidate grids and evaluation rules fixed. Rich features retain the three compact features and add nine behavioural measures. The research addresses:

1. The effect of representation on customer assignments, geometric quality and repeatability for K Means and CLASSIX.
2. The effect on CLASSIX provenance, ExKMC accuracy at a given rule budget, and compliance with operational checks.
3. Sensitivity to feature removal, reduced dimension, numerical changes and alternative session definitions.

The contribution links representation, clustering, explanation and operational assessment in one paired design. It separates within-space repeatability from between-space agreement and intrinsic provenance from post hoc approximation. Mathematical analysis explains CLASSIX projection pruning and local sufficient conditions for an unchanged distance-merging fit. It does not prove general CLASSIX stability or identify objectively correct customer types.

## 1.3 Dissertation structure

Chapter 2 positions the study in the literature; Chapter 3 defines its data, methods and mathematical scope. Chapter 4 begins with labelled benchmarks, including DBSCAN, which joins dense regions and can leave sparse observations unassigned, and Ward's method, which merges groups to limit increases in within-group squared dispersion (Ester et al., 1996; Ward, 1963). Labels evaluate benchmark results but do not select parameters. Customer comparisons then address the three research questions without treating either fitted partition as ground truth. Chapter 5 draws a bounded conclusion.

# Chapter 2

## Literature Review

### 2.1 Representation in customer segmentation

An event log records timestamped actions, such as viewing an item, adding it to a cart or purchasing it. Feature construction determines which similarities a clustering algorithm can observe. The choice would be neutral only if it left the relevant grouping unchanged.

Clickstream research derives behavioural segments from constraints on large, noisy action sequences (Dextras-Romagnino and Munzner, 2019). Object-centred event-log research constructs vector profiles for clustering and pattern analysis (Faria Junior et al., 2023), while customer-behaviour studies select features for particular decision tasks (Abdi and Abolmakarem, 2019). These application-specific choices motivate examining representation within the model rather than assuming one universally appropriate customer description.

### 2.2 Cluster structure and validation

Internal validation measures feature geometry; external validation compares assignments with supplied labels; repeated fitting tests recurrence under specified input changes. These are distinct forms of evidence. Clusterability theory supplies incompatible notions of recoverable structure (Ackerman and Ben-David, 2009), and empirical measures depend on distribution, dimension and the structure sought (Adolfsson, Ackerman and Brownstein, 2019). A high internal score therefore supports a partition within a given space, not representation-independent customer types. Dimension reduction also changes geometry and may weaken the connection to named attributes (Bertsimas, Orfanoudaki and Wiberg, 2021).

## 2.3 Explainable clustering

Three explanation designs have different targets. Intrinsic methods expose the computation producing their own partition: CLASSIX records sorted aggregation groups, starting points and final assignments (Chen and Güttel, 2024). Explainability-first methods incorporate a simple description into cluster formation: ICOT optimises tree partitions, while XClusters balances distortion against tree size and accuracy (Bertsimas, Orfanoudaki and Wiberg, 2021; Hwang and Whang, 2023). Post hoc methods approximate a reference partition with rules. Changing an explanation can consequently change the clustering in the second design, while the third evaluates an already fitted reference.

Table 2.1 Comparison of explanation designs in the reviewed literature

| Method    | Target                | Explanation stage | Form                       | Complexity control                | Status     |
|-----------|-----------------------|-------------------|----------------------------|-----------------------------------|------------|
| CLASSIX   | Own partition         | Intrinsic         | Groups and starting points | Radius and minimum size           | Journal    |
| IMM       | K Means objective     | Tree construction | Threshold tree             | One leaf per cluster              | Conference |
| ExKMC     | K Means labels        | Post hoc          | Expanded tree              | Leaf budget                       | Preprint   |
| CREAM     | Data or model outputs | During induction  | Cubes and tree rules       | Maximum depth and error threshold | Conference |
| ICOT      | Own partition         | Joint             | Optimised tree             | Validation objective and tree     | Journal    |
| XClusters | Joint objective       | Joint             | Decision tree              | Distortion and node tradeoff      | Conference |

## 2.4 Explanation complexity and fidelity

IMM constructs threshold trees and bounds the additional K Means or K Medians cost imposed by that restriction (Moshkovitz et al., 2020). Such simplicity can incur unavoidable objective loss depending on cluster count and dimension (Laber and Murtinho, 2021). A distance-cost guarantee is not a guarantee of label agreement.

Methods that remove outliers before explaining a clustering also change the population being explained (Bandyapadhyay et al., 2022).

CREAM constructs rectangular regions through tree induction, using clustering during construction and permitting model-output approximation in supervised settings. Maximum depth and predictive-error thresholds control splitting (Sabbatini and Calejari, 2023). These controls differ from CLASSIX radius and group counts, or the distortion/tree tradeoffs of ICOT and XClusters. None directly measures human comprehension.

ExKMC expands a tree beyond one leaf per K Means cluster. Its cost analysis motivates varying the leaf budget but does not guarantee held-out label agreement; the source used here is the authors' arXiv preprint (Frost, Moshkovitz and Rashtchian, 2020). Here *fidelity* means exact agreement between tree predictions and held-out K Means assignments. It measures approximation, not whether the reference clustering is correct. XAI research likewise shows why fidelity requires an explicit target and calculation (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025).

## 2.5 Research gap and positioning

Operational usefulness is another evidential boundary. Named rules may be queryable and provenance may identify representatives, without establishing effective interventions or improved revenue. Segmentifier supports behavioural refinement and export, but does not validate clustering's commercial effect (Dextras-Romagnino and Munzner, 2019). Objective-cost guarantees similarly do not establish adequate segment size or feature coverage.

The reviewed methods provide useful constructions and guarantees but do not establish that adding behavioural features improves all these properties together. This dissertation examines the link between representation, partition change, CLASSIX provenance, ExKMC fidelity and auditable addressability for the same purchasers. These remain separate assessments rather than components of one explainability score.

# Chapter 3

## Data and Methodology

This chapter explains how the RetailRocket event data are converted into two customer representations and evaluated under matched analytical conditions. It also distinguishes the mathematical properties of CLASSIX from the empirical robustness checks conducted in this study.

### 3.1 Research design

Compact and rich representations describe the same 11,719 purchasers using three purchase variables and twelve variables respectively. The nine additions measure activity, conversion, temporal engagement and category behaviour.

#### 3.1.1 Data source

The empirical comparison uses the RetailRocket recommender system dataset, an event log released under the Creative Commons Attribution NonCommercial ShareAlike licence (Retailrocket, no date). The data used in the analysis contain 2,756,101 event rows, two item property files with 20,275,902 rows in total, and a category tree with 1,669 nodes. Events record anonymous visitor, time, item and action; purchase events have transaction identifiers but no price or revenue. Counts measure activity, not expenditure or product quantity.

Visitors with at least one transaction event form the cohort: 11,719 purchasers and 230,678 event rows, identically ordered in both representations. The observation snapshot is one day after the final recorded event, which gives 19 September 2015 at 02.59.47 UTC. Purchase recency is measured from this snapshot rather than from each customer's final browsing action. Stable ordering by visitor, timestamp and original row position resolves

equal timestamps without making an arbitrary event sequence depend on file reading behaviour.

### 3.1.2 Feature engineering

Table 3.1 defines all twelve features. Purchase occasion count uses distinct transaction identifiers, whereas purchase event count uses rows; several rows can share an identifier. The three compact variables enter the rich table without recalculation.

Table 3.1 Recorded feature contract for the two customer representations

| Feature                 | Block               | Operational definition                                       |
|-------------------------|---------------------|--------------------------------------------------------------|
| Purchase recency days   | Compact transaction | Days from the latest transaction event to the snapshot       |
| Purchase occasion count | Compact transaction | Distinct nonempty transaction identifiers                    |
| Purchase event count    | Compact transaction | Transaction event rows                                       |
| Total events            | Activity            | All event rows for the visitor                               |
| Active days             | Activity            | UTC calendar days containing an event                        |
| Cart per view           | Conversion          | Add to cart events divided by view events                    |
| Transaction per view    | Conversion          | Transaction events divided by view events                    |
| Activity span days      | Temporal engagement | Days between first and last event                            |
| Mean interevent hours   | Temporal engagement | Mean gap between adjacent events                             |
| Session count 30m       | Temporal engagement | Sessions separated by more than thirty minutes               |
| Distinct top categories | Category behaviour  | Distinct root categories among known joins                   |
| Top category share      | Category behaviour  | Largest root category count divided by known category events |

Category variables require a time-valid join because item properties change during the observation period. Each event receives the latest category property that was available at or before its timestamp. The category tree then maps that category to its root. This



procedure covers 2,099,173 event rows, which is 76.16 per cent of all events. Unknown joins are excluded from category denominators rather than treated as a category. A total of 1,168 purchasers have no event with a known category. Their two category features are set to zero, while a separate coverage indicator records the missing information. This indicator is used for data checks and is not included among the twelve clustering variables. The missing category records prevent these variables from being interpreted as complete purchase preferences.

Ratio variables also use explicit denominator rules. Cart per view and transaction per view equal zero when a visitor has no recorded view event. This condition affects 428 purchasers and is retained in the feature audit. A session starts with a visitor's first event or after an inactivity gap greater than thirty minutes. Alternative gaps of fifteen and sixty minutes are reserved for sensitivity analysis.

Nonnegative counts and time variables receive a logarithm after adding one. Ratios and shares remain on their original scale. Each representation is then standardised feature by feature using its own fitted means and standard deviations. Full-data clustering fits these transformations on the complete cohort. Sample splitting and resampling fit them only on the relevant training or subsample rows. Principal component analysis (PCA), which replaces features with orthogonal linear combinations ordered by variance, is applied only to a separately standardised copy of the rich representation and never supplies features to rule explanations or cluster descriptions.

Feature checks verify identical purchaser identifiers, order and common purchase values across both tables. Raw-file hashes identify the source inputs; reconstructing temporal joins requires those originals. The feature contract records equal-timestamp, missing-category and zero-denominator handling. These checks isolate additional features from changes in population or purchase summaries.

## 3.2 Analytical methods and mathematical scope

The comparison uses one recorded customer cohort, feature contract, set of candidate grids, deployment constraints and evaluation measures. K Means provides a centre-based reference partition that can be approximated by ExKMC rules. CLASSIX provides a separate clustering result with native aggregation provenance. ExKMC is used here to approximate a fitted K Means reference, not as an independently selected customer clustering method; no composite score ranks it against CLASSIX. The same change in represented information is traced through each partition and its corresponding explanation.

### 3.2.1 Clustering candidates and selection

K Means: a partition around centres. For  $n$  observations  $x_i \in \mathbb{R}^d$  ( $i = 1, \dots, n$ ),  $d$  denotes the number of features. K Means seeks assignments  $z_i \in \{1, \dots, K\}$  and centres  $\mu_j \in \mathbb{R}^d$  ( $j = 1, \dots, K$ ) that minimise

$$\min_{z_1, \dots, z_n, \mu_1, \dots, \mu_K} \sum_{i=1}^n \|x_i - \mu_{z_i}\|_2^2.$$

The Euclidean norm  $\|\cdot\|_2$  measures straight-line distance; squaring it gives more weight to large deviations. The usual alternating procedure assigns each observation to its nearest centre, then replaces each centre by the mean of its assigned observations (MacQueen, 1967; scikit-learn developers, 2025). Each step does not increase the objective, but the fitted solution need not be the global minimum. The nearest-centre regions are Voronoi cells: each contains the points closer to its centre than to any other. k-means++ initialisation favours points far from already chosen centres using squared-distance sampling; repeated initialisations reduce dependence on one starting configuration (Arthur and Vassilvitskii, 2007).

Feature scaling matters because every coordinate contributes to squared distance. Adding nine standardised behavioural dimensions changes that sum even when the original three coordinates are unchanged. Centres and their nearest-centre assignments can consequently change for the same purchasers. Standardisation removes measurement-unit differences; it does not make the choice of features inconsequential.

CLASSIX: local groups followed by merging. CLASSIX first aggregates observations into small groups and then joins groups into clusters (Chen and Güttel, 2024). The study supplies standardised observations. In the version 1.5.1 Euclidean PCA branch, the package additionally centres them and divides by the median centred Euclidean norm (using one if that scale is zero). The radius below is in these processed coordinates, not in raw feature units (Chen and Güttel, 2026).

The *principal sorting direction* is a unit vector along which the processed observations have maximum variance. An observation's *projected score* is its scalar coordinate along that direction. After sorting by these scores, the first unassigned observation becomes a *starting point*. Later unassigned observations within the *aggregation radius* of that point join its *aggregation group*. Points already assigned to an earlier group are skipped. The next unassigned point starts another group, until every observation belongs to one. A starting point is therefore an observed representative, not a fitted centroid; a group is an intermediate object, not necessarily a final cluster.

For *distance merging*, groups are linked when their starting points are within the radius multiplied by the merge scale, and connected groups share a cluster. For *density merging*, the criterion uses counts per volume in overlapping radius balls: version 1.5.1 links a pair when the intersection density is at least one of the two ball densities. It thus tests local overlap density rather than only centre separation. Minimum cluster size controls the subsequent treatment of small merged components. The recorded distance branch reassigns groups from undersized components to a nearest eligible component. In the density branch, disabling post allocation leaves rejected observations labelled  $-1$ , or *noise*. Noise denotes absence from a retained cluster under that configuration, not an invalid customer record. Small-group merging and final outlier allocation are distinct options (Chen and Güttel, 2026).

The resulting *provenance* connects each observation to its aggregation group, starting point and final label. This construction record is the reason for using CLASSIX here. It explains its own grouping process, whereas a post hoc tree approximates an already fitted reference. Neither representation is preferred in advance.

K Means uses k-means++ seeding, twenty initialisations and the fixed seed 20260801 (MacQueen, 1967; Arthur and Vassilvitskii, 2007). The number of clusters ranges from 2 to 12. CLASSIX uses Euclidean distance and sorting along the first principal direction. Radius ranges from 0.25 to 1.75 in increments of 0.05. Minimum cluster size takes values 1, 5 and 10, while group merging uses either distance or density. The merge scale is 1.5, tiny groups can be merged, and post allocation is disabled. Each representation has 11 K Means candidates and 186 CLASSIX candidates. A recorded boundary diagnostic triggered a symmetric radius extension from 1.80 to 2.25 for both representations. The extension tests grid sensitivity and does not replace the primary selection.

Each method and representation reports an unconstrained silhouette optimum and a deployment-acceptable result. Silhouette compares an observation's mean distance to its own cluster with its smallest mean distance to another cluster; larger values indicate better separation relative to cohesion. Davies–Bouldin averages each cluster's worst ratio of within-cluster spread to centre separation, so smaller values are preferred. Calinski–Harabasz compares between-cluster with within-cluster squared dispersion, adjusted for sample and cluster counts, and larger values are preferred. These are different geometric criteria, not competing estimates of ground truth. A candidate is internally valid only when at least two nonnoise clusters remain and silhouette, Davies Bouldin and Calinski Harabasz indices are finite (Rousseeuw, 1987; Davies and Bouldin, 1979; Caliński and Harabasz, 1974). Deployment acceptability further requires 2 to 12 nonnoise clusters, no more than 20 per cent noise and no single nonnoise cluster larger than 95 per cent of nonnoise observations. Selection maximises silhouette, then prefers lower Davies Bouldin, higher Calinski Harabasz, a smaller largest cluster share, less noise,

fewer clusters and finally the lexical parameter record. Runtime is measured but never selects a model. Keeping the unconstrained result reveals when an attractive internal score is produced by a partition that is unsuitable under the stated operational constraints.

Internal indices exclude observations labelled as noise, so CLASSIX index comparisons refer to each partition's nonnoise subset. Cross-representation partition comparisons retain every purchaser because a noise assignment is itself a difference in assignment. Fixed K comparisons use the same value from 2 to 12. Fixed CLASSIX comparisons use the same radius, minimum size and merge rule. Independently selected deployment partitions are compared by adjusted Rand index, arithmetic-mean normalised mutual information and variation of information (Hubert and Arabie, 1985; Meilă, 2007). ARI measures agreement in whether pairs of observations share a cluster, adjusted for the agreement expected under its fixed-size random-partition model. One means identical partitions up to labels; zero is that chance-adjusted reference, and negative values are possible. It can compare two fitted partitions without ground truth: here it measures agreement, not accuracy. Arithmetic-mean NMI divides shared label information by the mean entropy of the two partitions; VI is their sum of conditional entropies, measured in nats with natural logarithms. Higher NMI and lower VI indicate closer agreement (scikit-learn developers, 2025; Meilă, 2007). A Hungarian match finds the one-to-one relabelling that maximises overlapping assignments for transition tables (Kuhn, 1955). The resulting migration share describes changed assignments and does not imply that numeric labels identify persistent customer types.

### 3.2.2 Resampling and explanation evaluation

Fixed-parameter overlap-resampling stability uses ten pairs of independent 80 per cent samples without replacement. The same seed pairs are used for both representations and methods. Each sample refits its transformations and clustering with the full-data deployment parameters. Adjusted Rand index is calculated only on purchaser identifiers present in both members of a pair. The statistic measures sensitivity to sampled customers and, for K Means, initialisation, conditional on the selected configuration. It does not measure stability of model selection, calendar time or an external cohort.

The CLASSIX explanation retains public aggregation group assignments, starting point indices, final labels and cluster sizes. Merge membership is reconstructed from public group and final assignments because version 1.5.1 does not expose a documented complete edge list. Native aggregation groups use sorted rows, so the export restores input order with the saved inverse permutation before linking groups to visitors. Every exported provenance label is checked against the selected main result for the same visitor, and every representative must belong to its reported group. The trace is an explanation of the

fitted CLASSIX partition and does not claim that a group boundary is a human validated customer rule.

ExKMC approximates K Means labels under five deterministic 80 per cent training and 20 per cent test splits. The primary full cohort selection gives four K Means clusters in both representations, so K is fixed at four before splitting. Log transformations, standardisation, the K Means reference and the ExKMC tree are fitted using training rows only. The test target is the label returned by the fitted K Means predictor. This estimates explanation fidelity conditional on the primary K choice rather than nested selection of K. A leaf is a terminal decision-tree region; the leaf budget caps the number of such regions, and tree depth counts tests on the longest root-to-leaf path. The leaf budget ranges from K to four times K. The main comparison fixes the budget at K, while the wider range measures the fidelity and complexity tradeoff. A supplementary analysis selects K from 2 to 12 using each training split alone. Fidelity is the proportion of exact label agreements. Complexity is reported through effective leaves, depth, path conditions and features. Rule thresholds are exported in both standardised and original units.

Technical addressability means satisfying the study’s specified size, coverage and named-feature distinctiveness requirements. It is evaluated for selected CLASSIX and K Means clusters through a documented audit rather than subjective cluster names. A nonnoise cluster must contain at least 50 customers and one per cent of nonnoise customers, retain at least 80 per cent category coverage, and differ from the global median by at least half an interquartile range on two to four named features. Passing this audit shows that a cluster can be expressed as a limited set of query conditions with adequate coverage. It does not demonstrate intervention effectiveness, revenue change or human comprehension.

### 3.2.3 Projection filtering and complexity

CLASSIX can rule out some candidate neighbours before computing their full Euclidean distance. The purpose of this section is to justify that computational shortcut and distinguish its theoretical cost from recorded performance. Proposition 3.1 restates the projection argument underlying CLASSIX aggregation (Chen and Güttel, 2024); it is not a new clustering objective.

Let  $x_i \in \mathbb{R}^d$  be processed observation  $i$ , where  $d$  is the number of coordinates, and let  $u$  be a unit sorting direction. Define its score by  $s_i = u^\top x_i$ . A starting point is denoted by  $c$ , with score  $s(c) = u^\top c$ , and  $r > 0$  is the aggregation radius in the same processed coordinates. Scores are traversed in nondecreasing order.

**Proposition 3.1.** *If a later point satisfies  $s_i - s(c) > r$ , then it cannot satisfy  $\|x_i - c\|_2 \leq r$ .*

*Proof.* The Cauchy–Schwarz inequality bounds an inner product by the product of the two vector lengths. Since the projection direction has length one,

$$|s_i - s(c)| = |u^\top(x_i - c)| \leq \|u\|_2 \|x_i - c\|_2 = \|x_i - c\|_2.$$

Thus a projected separation greater than  $r$  implies a full distance greater than  $r$ . □

In words, points already too far apart along one unit direction cannot be close in the full space. Once a sorted score exceeds the window, every later score does too, so the scan for this starting point can stop. The converse fails: a point inside the score window may be far away in other coordinates and still needs an exact distance test. Equality remains eligible in the implementation. The argument is exact in real arithmetic; it does not remove floating-point boundary effects or the order dependence of greedy aggregation.

Let  $n$  be the number of observations and  $q$  the number of resulting aggregation groups. At most  $n(n - 1)/2$  distinct later-point pairs can be tested during aggregation. Distance merging has at most  $q(q - 1)/2$  distinct starting-point pairs. Each full distance involves  $d$  coordinates, giving the conservative distance-arithmetic bound

$$O((n^2 + q^2)d).$$

This is a worst-case bound for those distance operations, not an average runtime law. Principal-direction estimation, scalar sorting and bookkeeping have additional costs; small-component reassignment can add a quadratic group-distance pass. The bound on distinct pairs need not equal the number of vectorised arithmetic operations, which may include self-comparisons.

The saved package counter counts retained aggregation distance comparisons on the tested inputs; elapsed runtime also includes preprocessing, merging and implementation overhead. Neither an observed small counter nor a short runtime proves linear complexity. The distinction matters here because the grouping trace arises during clustering itself. Its computational mechanism differs from the later construction of an ExKMC tree.

### 3.2.4 A local stability certificate

Section 4.4.2 changes numerical feature values slightly and measures whether CLASSIX assignments change. One possible cause of change is rotation of the principal sorting direction. This section asks a narrower mathematical question: under what sufficient conditions do the finite decisions of a distance-merging fit remain unchanged? It separates stability of a direction from stability of a complete partition.

Write  $S$  for the covariance matrix of the centred, fully processed observations, and  $\lambda_1 \geq \lambda_2$  for its two largest eigenvalues. The *eigengap*  $\gamma = \lambda_1 - \lambda_2 > 0$  measures how clearly the leading direction is separated from the next. A perturbation is a change of the numerical input: it gives processed observations  $\hat{x}_i$  and covariance  $S + \Delta S$ , where  $\Delta S$  is the covariance change. Let  $u$  and  $\hat{u}$  be unit leading eigenvectors before and after this change. Choose their relative signs so that  $u^\top \hat{u} \geq 0$ .

Davis–Kahan bounds the rotation of an eigenspace when its matrix changes (Davis and Kahan, 1970). Using the one-dimensional form of Yu et al. (2015, Corollary 1), and restricting attention to  $2\|\Delta S\|_2 < \gamma$ , gives

$$\sin \angle(\hat{u}, u) \leq \frac{2\|\Delta S\|_2}{\gamma}.$$

Here the matrix norm is the spectral norm (the largest singular value); the angle is the acute angle between the two directions. A larger eigengap permits a smaller rotation bound for the same covariance change. If the leading directions are nearly tied, this argument gives little protection.

Let  $\varepsilon_x$  bound each processed displacement  $\|\hat{x}_i - x_i\|_2$ , and let  $B$  bound every original processed norm  $\|x_i\|_2$ . The sign-aligned unit vectors obey  $\|\hat{u} - u\|_2 \leq \sqrt{2} \sin \angle(\hat{u}, u)$ . Consequently,

$$\begin{aligned} |\hat{u}^\top \hat{x}_i - u^\top x_i| &\leq \|\hat{x}_i - x_i\|_2 + \|\hat{u} - u\|_2 \|x_i\|_2 \\ &\leq \varepsilon_x + \sqrt{2}B \frac{2\|\Delta S\|_2}{\gamma} =: b_s. \end{aligned}$$

Thus  $b_s$  is a maximum score movement, accounting for movement of both the point and the direction. The input to this bound is after all preprocessing: changes of fitted centring and scale must be included in  $\varepsilon_x$  and  $\Delta S$ . A Gaussian noise standard deviation is not itself a bound on every processed displacement.

The remaining conditions protect individual decisions from crossing thresholds. Define  $g_s$  as the minimum adjacent gap in the original sorted scores. Define  $g_o = |u^\top x_1|$  for the first input row used by version 1.5.1’s score-sign convention. Define  $g_w$  as the smallest absolute distance of any relevant score difference from its active search-window threshold, including both aggregation radius  $r$  and distance-merge radius  $1.5r$ . Let  $g_r$  be the smallest absolute margin of an aggregation distance from  $r$ , and  $g_m$  the corresponding distance-merge margin from  $1.5r$ . These minima refer to the decisions traversed by the original fit; they must be positive for the certificate below.

Let  $b_r$  and  $b_m$  bound changes in the aggregation and merge distance tests respectively. With unchanged radii in fully processed coordinates, both may conservatively be set to

$2\varepsilon_x$ , by the triangle inequality. If thresholds change, their changes must also be added to the relevant bounds, including the score-window bound.

*Proposition 3.2. For a fixed distance-merging configuration and unchanged processed thresholds, the same partition is produced if*

$$g_s > 2b_s, \quad g_o > b_s, \quad g_w > 2b_s, \quad g_r > b_r, \quad g_m > b_m,$$

*provided that any later nearest-eligible-group choices and tie decisions in small-component handling are also unchanged. The input identities, feature dimension, minimum-size rule and implementation branch are fixed. The claim is in exact arithmetic, or with numerical error included in the bounds.*

*Proof.* The first condition preserves the sorted order: two scores can approach each other by at most  $2b_s$ . The second preserves the orientation anchor’s sign, so the package’s orientation agrees with the sign alignment used in the score bound. The third preserves the relevant search windows. Proceeding through the original traversal, the next unassigned starting-point identity is therefore the same; the fourth condition preserves its accept/reject decisions. Induction gives the same aggregation memberships and representative identities, although their coordinates may move. The third and fifth conditions then preserve the distance-merge links, hence their connected components and sizes. Minimum-size eligibility is consequently unchanged. The additional assumption on nearest-group choices and ties preserves any subsequent reallocation, giving the same final partition up to label names.  $\square$

Each inequality says that a decision’s distance from its boundary must exceed the largest allowed movement. These requirements are deliberately conservative: failure of one does not imply a changed partition. The certificate is local, conditional and sufficient, not necessary. It is not a general proof of CLASSIX stability, and the recorded experiments do not establish that all these margins hold. Density merging additionally depends on overlap counts and volumes, which these conditions do not control. In particular, the selected rich CLASSIX fit uses density merging and is not certified by Proposition 3.2. Its empirical perturbation results in Section 4.4.2 are complementary observations, not an application of this proposition.

### 3.2.5 Benchmark and robustness designs

Nine labelled datasets provide an algorithm-level benchmark for CLASSIX, K Means, DBSCAN and Ward (Ester et al., 1996; Ward, 1963). Candidate selection uses only internal indices and deployment guards. Reference labels are reserved for ARI and NMI



evaluation. K Means results average five fixed seeds before parameter selection. Ward, DBSCAN and CLASSIX store one structural row per candidate, while repeated fits audit label identity and runtime on the recorded environment. This empirical identity check is not a global determinism theorem.

The rich representation undergoes twelve leave-one-feature-out fits. Principal component analysis (PCA) rotates the standardised features into orthogonal directions ordered by explained variance. Here it retains at least 90 per cent of standardised variance and is compared with the named space on structure and stability, never on rule quality (Pearson, 1901; Hotelling, 1933). The CLASSIX perturbation check adds independent zero-mean Gaussian noise to each standardised feature entry, then refits with the selected parameters and compares labels with the unperturbed fit. It uses noise standard deviations 0, 0.000001, 0.0001, 0.001 and 0.01 with ten seeds at every nonzero level. Session counts are rebuilt with gaps of 15, 30 and 60 minutes. A negative control appends independently permuted rich-only dimensions to unchanged compact features. These diagnostics retain the primary selected configurations unless a section explicitly reports complete feature reselection.

# Chapter 4

## Results and Analysis

The results follow the evidence chain established in Chapter 3. A benchmark first checks the implementations and selection procedure. The customer analysis then examines fitted partitions, explanation and addressability before sensitivity tests probe the main observations.

### 4.1 Algorithm benchmark

Nine labelled datasets (six shape and three multivariate datasets) check the algorithms under label-free selection before the customer analysis.

No method dominates the benchmark. K Means has the highest mean ARI across the nine selected results at 0.597, followed by Ward at 0.575, DBSCAN at 0.509 and CLASSIX at 0.508. CLASSIX and Ward share a mean rank of 2.44, while K Means has 2.39 and DBSCAN has 2.72. These averages conceal marked dependence on shape. CLASSIX recovers the Jain reference exactly but reaches only 0.264 ARI on R15. DBSCAN recovers Spiral exactly and performs best on Aggregation and Compound, yet its selected Wine candidate has no valid internal score. K Means is strongest on Seeds and Wine, while Ward is strongest on R15.

Runtime depends on implementation and execution conditions, not a universal speed ordering. The benchmark checks the shared evaluation pipeline; the customer conclusion rests on paired representation evidence.

### 4.2 Representation and customer cluster structure

Table 4.2 compares independently selected deployment-acceptable fits. K Means selects four clusters in each space, but their agreement is 0.014 ARI. CLASSIX selects two

Table 4.1 Highest external ARI after label free selection on each benchmark dataset

| Dataset     | Method              | ARI   |
|-------------|---------------------|-------|
| Aggregation | DBSCAN              | 0.910 |
| Compound    | DBSCAN              | 0.826 |
| Iris        | CLASSIX and K Means | 0.568 |
| Jain        | CLASSIX             | 1.000 |
| Pathbased   | CLASSIX             | 0.497 |
| R15         | Ward                | 0.804 |
| Seeds       | K Means             | 0.481 |
| Spiral      | DBSCAN              | 1.000 |
| Wine        | K Means             | 0.897 |

compact and three rich nonnoise clusters; its 218 noise assignments are purchasers not allocated to a retained cluster under density merging. Cross-representation ARI is 0.068. These values compare two fitted assignments on the same purchasers, not accuracy against known customer types.

Table 4.2 Deployment acceptable clustering results on the shared purchaser cohort

| Representation | Method  | Clusters | Noise | Largest share | Silhouette | Stability |
|----------------|---------|----------|-------|---------------|------------|-----------|
| Compact        | CLASSIX | 2        | 0     | 0.909         | 0.595      | 0.985     |
| Rich           | CLASSIX | 3        | 218   | 0.898         | 0.291      | 0.965     |
| Compact        | K Means | 4        | 0     | 0.664         | 0.560      | 0.988     |
| Rich           | K Means | 4        | 0     | 0.579         | 0.351      | 0.995     |

Internal indices disagree because they weight separation and dispersion differently. Compact CLASSIX has silhouette 0.595, Davies Bouldin 1.057 and Calinski Harabasz 4285.5; rich values are 0.291, 0.859 and 1516.9. Davies Bouldin favours rich, while the other indices favour compact; each uses its partition's nonnoise subset. Compact K Means has silhouette 0.560 and Calinski Harabasz 10459.1, compared with 0.351 and 3923.4 for rich, while Davies Bouldin rises from 0.775 to 1.040. No criterion establishes a representation-independent quality ranking.

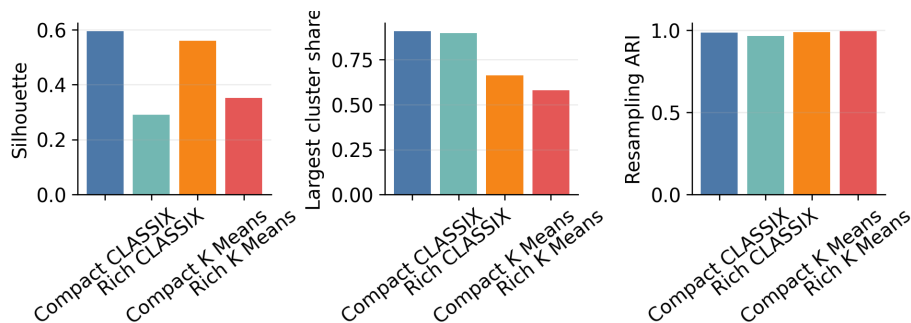


Figure 4.1 Internal structure and resampling stability for the selected customer partitions

Silhouette alone selects nearly undivided populations. Compact CLASSIX reaches 0.899 with 11,698 of 11,719 customers in one cluster; rich CLASSIX reaches 0.792 with 11,716 in one cluster and three across two tiny clusters. Compact K Means reaches 0.850 with 11,593 in one of two clusters. Isolating distant outliers can give the majority high silhouette without useful subdivision. All three fail the 95 per cent maximum-share rule, an operational choice rather than a mathematical law. Reporting both optima exposes this tradeoff. Only rich K Means has identical unconstrained and deployment selections.

Low agreement is not caused only by choosing different numbers of clusters. At every fixed K from 2 to 12, the K Means ARI between compact and rich representations remains between 0.012 and 0.113. At the common selected value of four it is 0.014, with NMI 0.089 and variation of information 1.691 nats. Under matched labels, 54.61 per cent of customers move to a different K Means cluster. Among 41 CLASSIX settings that are deployment acceptable in both spaces, matched-parameter ARI ranges from  $-0.051$  to 0.392 with a median of  $-0.023$ . The independently selected CLASSIX partitions have NMI 0.054 and variation of information 0.693 nats, with 18.66 per cent assigned to a different matched cluster or noise state. The lower CLASSIX migration share partly reflects the dominance of one large cluster in both representations and should not be read as stronger substantive agreement.

Cluster sizes make the changes more concrete. Compact K Means contains 7,784, 2,494, 1,344 and 97 customers. Rich K Means contains 6,790, 3,607, 1,156 and 166. Its 166 customer cluster has a median of 338.5 total events, 28 sessions and 12 top categories, whereas the full cohort medians are 6 events, 2 sessions and 1 top category. Another rich cluster of 1,156 customers has no known category at its median and a purchase recency median of 127.4 days. These are descriptive contrasts in observed features, not customer identities or causal segments.

CLASSIX is more imbalanced: compact cluster sizes are 10,656 and 1,063; rich sizes are 10,331, 1,137 and 33, plus 218 noise assignments. The 33-customer group has medians of 869 total events, 36 sessions and 14 top categories. Rich features thus expose unusual activity and missing-category groups, but operational assessment below shows why these need not be usable segments.

Additional standardised coordinates change distances, K Means centres and CLASSIX sorting/neighbourhood decisions. Resampling retains one representation while changing fitted purchasers; cross-space comparison changes geometry. High repeatability and low cross-space agreement are therefore compatible, without identifying a causal share attributable to individual features.

Mean overlap-resampling ARI is 0.985/0.965 for compact/rich CLASSIX and 0.988/0.995 for K Means. The lowest of ten CLASSIX values is 0.881/0.941. Each selected

configuration usually reproduces its grouping when one fifth of customers is omitted. This is conditional repeatability, not proof of unique correctness or stable model selection.

### 4.3 Explanation complexity, fidelity and addressability

CLASSIX provenance describes its own construction; ExKMC fidelity evaluates approximation of K Means labels. Their values are not combined into one explanation score.

Compact CLASSIX produces 131 aggregation groups and two final clusters; rich produces 559 groups, three clusters and noise. Each customer trace records the visitor, group, representative and final label. Both exports reproduce selected labels exactly with ARI 1. The larger group count means finer computational provenance, not measured reading time or cognitive difficulty.

At four leaves, both representations have four rules, depth 3 and 2.25 conditions per rule. Compact training/test fidelity is 0.9466/0.9448; rich is 0.9639/0.9650. Across five splits, test fidelity ranges from 0.9416 to 0.9480 and 0.9599 to 0.9697 respectively. These are descriptive ranges, not inferential uncertainty; the training/test proximity gives no sign of a training-only gain.

Compact rules use purchase occasion count and purchase recency across all five splits. Rich rules use purchase occasion count, activity span and the number of distinct top categories. Thresholds are exported in original units after reversing the training scaler and logarithmic transform. Named thresholds support direct database style conditions, but they inherit the meaning and missingness limits of the constructed variables.

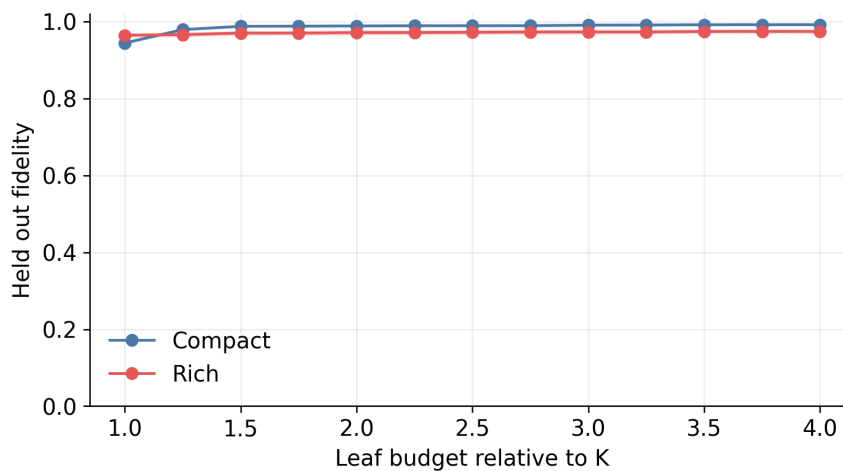


Figure 4.2 Held out ExKMC fidelity across the fixed leaf budget

At sixteen leaves, compact fidelity reaches 0.9924 with mean depth 6.6 and 4.84 conditions per rule; rich reaches 0.9747 with depth 6.4 and 4.74 conditions. Rich begins

higher at four leaves, but compact benefits more from additional cuts and overtakes it. Representation thus changes the observed budget–fidelity relationship, without establishing minimum possible complexity or human interpretability. Training-only K selection chooses two compact and four rich clusters in every split. Its compact fidelity is 0.9985 with two rules, but differing K excludes it from the main complexity comparison.

The addressability audit asks a further question beyond rule export. A cluster can exist without being well separated, be repeatable without having short rules, and have short rules without adequate size or feature coverage. Passing those operational checks still does not establish business value. These properties require separate evidence. Neither compact CLASSIX cluster passes because neither differs from the cohort median on at least two named features by the required half interquartile range. In the rich result, the 33-customer cluster is too small, the 1,137-customer cluster has zero category coverage, and the 10,331-customer cluster lacks two sufficiently distinctive median conditions. None of the five nonnoise CLASSIX clusters is classified as addressable. No compact K Means cluster passes. One of four rich K Means clusters passes all gates, while the others fail because they have too many distinguishing conditions or insufficient category coverage.

The one passing cluster meets a technical audit, not an outcome test. Queryability and coverage do not establish campaign effectiveness, revenue or human understanding.

## 4.4 Sensitivity and robustness

Fixed-parameter checks test local dependence; reselection repeats the candidate search and deployment rule. Permutation examines additional dimensions without aligned customer information.

### 4.4.1 Named features and reduced geometry

Fixed parameter deletion identifies category behaviour as the strongest local influence on the rich partitions. Removing distinct top categories reduces agreement with the full rich result to ARI 0.189 for CLASSIX and 0.516 for K Means. Removing top category share gives ARI 0.238 and 0.517. Every other CLASSIX deletion remains between 0.927 and 0.967. The corresponding K Means values remain between 0.855 and 0.991. Activity span and mean interevent time have the next largest effect on K Means, with ARI 0.874 and 0.855. Influence is unevenly distributed across the named variables, but category information alone does not generate a preferable segmentation.

The PCA diagnostic preserves 91.32 per cent of standardised rich variance in six components. It is a fixed-parameter local diagnostic rather than PCA-space model

selection. K Means in this reduced space agrees with the named rich partition at ARI 0.995 and has silhouette 0.390 compared with 0.351. CLASSIX retains three clusters, lowers noise from 218 to 71 customers and agrees with the named result at ARI 0.920. Its silhouette is 0.304 compared with 0.291. Mean resampling ARI is 0.995 for reduced K Means and 0.987 for reduced CLASSIX. Much of the rich partition therefore survives dimension reduction, without proving that redundancy is irrelevant or six components optimal. Rotated components lose the direct meaning needed for named-feature explanations.

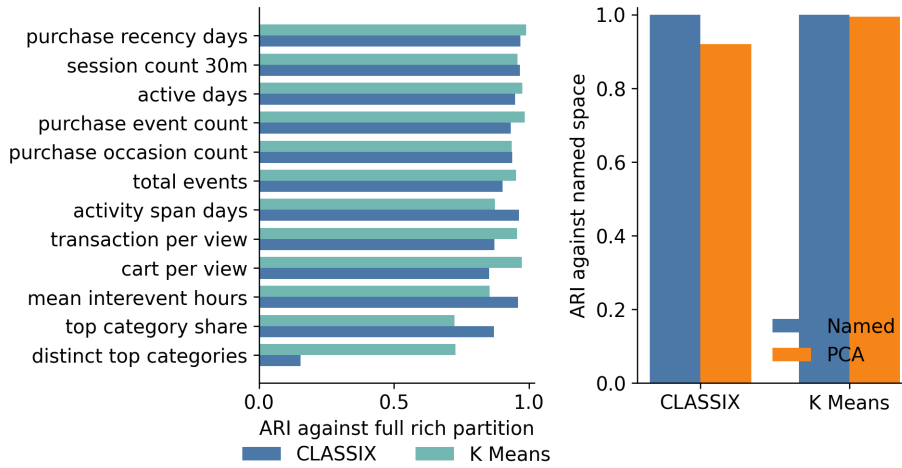


Figure 4.3 Reselected feature deletion and PCA diagnostics

Complete reselection confirms that category information creates the largest structural turn. All twenty two scenarios produce a deployment acceptable candidate. Removing distinct top categories changes CLASSIX to eleven clusters with ARI 0.153 against the full rich partition. K Means changes to two clusters with ARI 0.728. Removing the whole category block produces seven CLASSIX clusters with ARI 0.126 and two K Means clusters with ARI 0.729. The progressive endpoints reproduce the compact and rich primary partitions exactly. These 4,334 candidate fits show that the category effect persists when parameters and cluster counts are allowed to change.

CLASSIX sensitivity is distributed across the grid rather than concentrated at one radius. Deployment acceptable candidates occur at twenty three of thirty one compact radii from 0.35 to 1.45 and twenty rich radii from 0.30 to 1.25. Minimum size 1 produces no acceptable candidate in either representation. Minimum sizes 5 and 10 produce 23 and 32 acceptable compact candidates, compared with 19 and 27 rich candidates.

#### 4.4.2 Perturbation and temporal assumptions

Small Gaussian perturbations leave the selected CLASSIX partitions almost unchanged. Compact agreement remains ARI 1 through a noise standard deviation of 0.001. At 0.01

the mean ARI is 0.999 and the minimum is 0.996. Rich agreement is also exact through 0.0001. It falls to a mean of 0.995 at 0.001 and 0.985 at 0.01, with minima of 0.985 and 0.962. Rich assignments are more sensitive at the tested scales, without wholesale reassignment. These trials cannot certify all nearby inputs. In particular, the rich fit uses density merging, outside the distance-only certificate of Section 3.2.4.

Reconstructing sessions from raw events gives a similar qualified result. The selected thirty minute window reproduces both partitions exactly. A fifteen minute window gives ARI 0.976 for K Means and 0.971 for CLASSIX. A sixty minute window gives ARI 0.985 and 0.959. The four cluster K Means result and the three cluster CLASSIX result persist at both alternative windows, although the CLASSIX noise count changes from 218 to 220 and 239. The conclusions do not arise solely from choosing exactly thirty minutes, while the detailed assignments still depend on the session convention.

#### 4.4.3 Dimensionality control

The permutation control sets a stricter boundary on causal interpretation. The compact columns remain unchanged while the nine additional dimensions are independently permuted across customers. These columns preserve their marginal distributions but destroy their joint relationship with the compact representation and with each other. With the compact K Means setting held fixed, adding all nine permuted columns lowers agreement with the compact partition to ARI 0.017 and silhouette to 0.127. With the compact CLASSIX setting held fixed, adding even one permuted dimension produces a single cluster and ARI 0.

The single deterministic permutation control is an existence demonstration rather than a variance decomposition. Standardisation gives each added variable comparable weight, so random added dimensions can alter distances, principal directions and radius neighbourhoods in at least one realised order. The control does not estimate how much of the observed compact-to-rich difference is caused by dimensionality. Named feature deletion and PCA separately show that category and temporal information contribute reproducible structure in the observed rich representation.

### 4.5 Integrated discussion and validity limits

**Strongly supported findings** For both algorithms, representation materially changes assignments on the same purchasers, including under matched parameters. High within-space repeatability does not contradict between-space disagreement. Explanation also changes: rich CLASSIX has more aggregation groups, while ExKMC's relative fidelity reverses as its budget increases. The single passing rich K Means cluster



demonstrates why geometry, repeatability, rule approximation and operational addressability require separate evidence.

Conditionally supported findings    Deletion and reselection identify category dependence; PCA preserves much fitted structure, and tested numerical changes and session windows retain substantial agreement. These findings concern specific diagnostics, not all preprocessing choices. The permutation demonstrates a possible dimensionality effect without quantifying its causal share. Deployment cutoffs are operational choices; resampling holds selected parameters fixed; benchmark rankings need not transfer to purchasers. The local certificate requires unmeasured margins and excludes density merging.

Unresolved questions and evidence limits    One anonymous purchaser log cannot establish general customer types. Category joins cover 76.16 per cent of events; 1,168 purchasers have no known category, and zero ratios can reflect absent recorded views. No demographic, price, revenue or response data justify labels such as valuable or loyal.

Neither intrinsic provenance nor agreement with a K Means predictor measures human comprehension. Addressability checks size, coverage and named-feature distinctiveness, not commercial outcomes. Prospective interventions and user evaluation would be needed to establish those benefits. The contribution is the joint examination of separate clustering and explanation properties under representation change, not universal superiority of richer features.

# Chapter 5

## Conclusion

For the same 11,719 RetailRocket purchasers, compact and rich features produce substantially different K Means and CLASSIX assignments, including under matched parameters. High within-space resampling agreement therefore coexists with disagreement between representations.

Explanation properties also change. Rich ExKMC fidelity begins higher at four leaves, but compact exceeds it at sixteen. Rich CLASSIX has more aggregation groups, describing provenance detail rather than human difficulty. Only one rich K Means cluster passes the operational audit; accurate approximation and repeatable clustering alone do not establish addressability.

Sensitivity checks qualify these findings under feature deletion, dimension reduction, numerical perturbation and session changes. Projection analysis explains a valid computational shortcut; the local certificate supplies conditional distance-merging guarantees and does not certify the selected rich density-merging fit.

Representation is part of the empirical clustering model. Here richer behaviour changes assignments and explanation properties without consistently improving geometric quality or producing deployment-ready groups. Objectively correct customer types, human comprehension and realised business value remain unestablished.

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# Appendix A

## Technical Reproduction Map

The reported experimental results are reproduced from the distributed data with `python-msrc.reproduce`. The fixed sequence covers the benchmark, customer clustering, explanations, sensitivity analyses, summary tables and figures. The optional `--compile-pdf` flag also rebuilds the report after rerunning the experimental sequence. To compile only the manuscript while retaining saved results, run `xelatex-interaction=nonstopmode-halt-on-error-output-directory=buildmain.tex` three times from `dissertation/overleaf`. Environment versions and random seeds are recorded in `environment.yml` and `protocol/analysis_protocol.json`.

Primary clustering selections are stored in `results/retailrocket/selected_configurations.csv`. Customer level labels, transitions, profiles and resampling records remain separate so that every reported aggregate can be recalculated. ExKMC training selection, held out fidelity and exported rules are stored in distinct files. CLASSIX provenance includes an exact label audit against the selected result.

The sensitivity directory separates fixed parameter diagnostics from complete candidate reselection. PCA records that reduced components are not permitted for rules. The zero perturbation rows have ARI 1. Session windows are reconstructed from the distributed purchaser timestamp table. Permuted dimensions retain the compact columns exactly while destroying their joint customer alignment.

The distributed feature tables and purchaser event timestamps are sufficient for the reported experiments. The optional `--rebuild-features` flag reconstructs these tables when the four original RetailRocket files are available. The repository `README.md` records the execution sequence, output map and licence boundary.